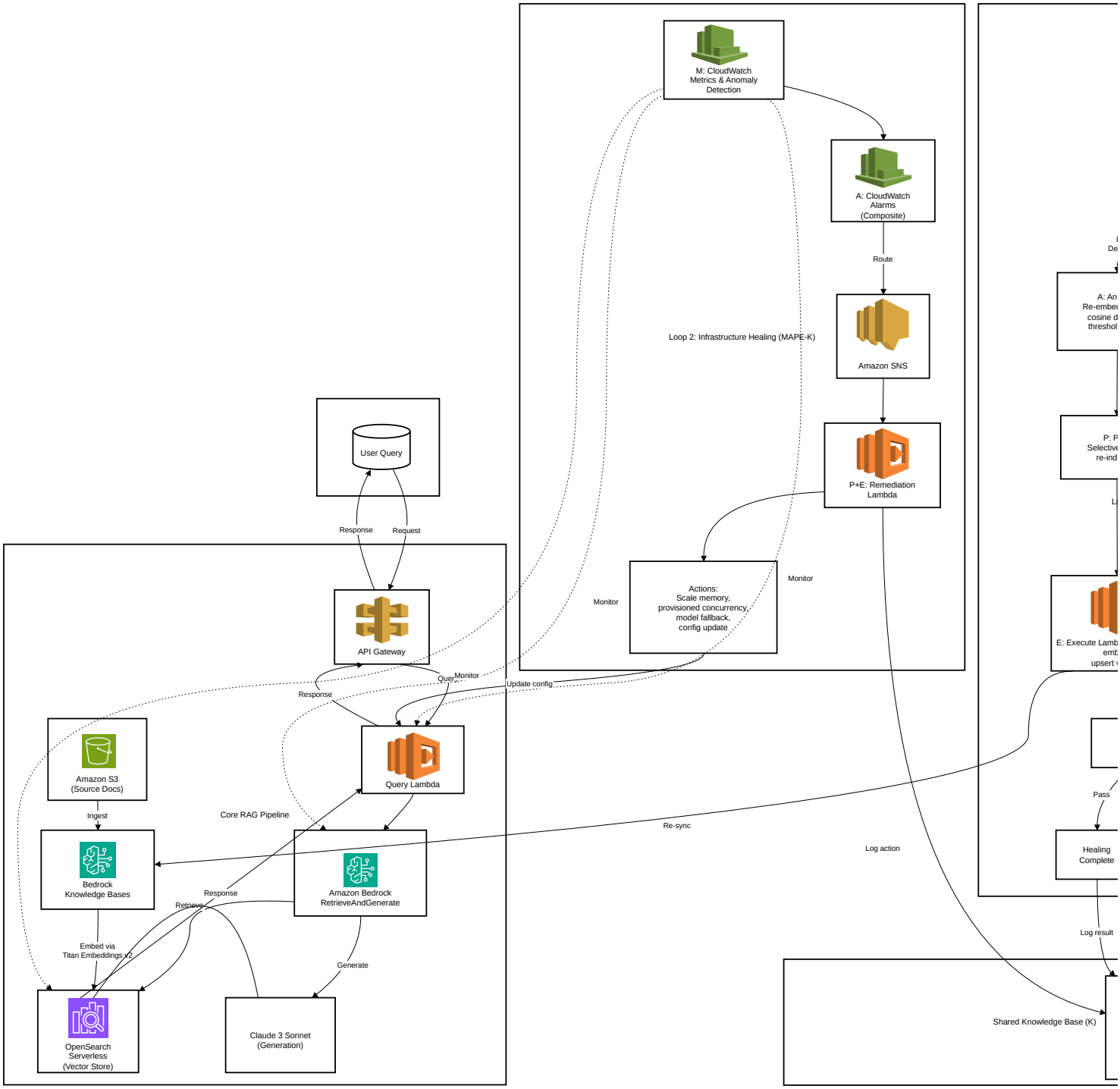
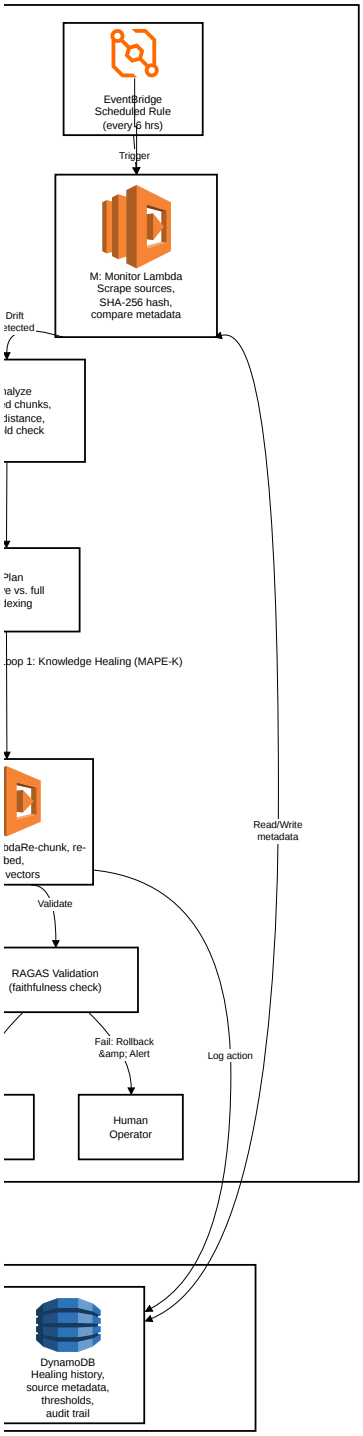

M: CloudWatch Metrics & Anomaly Detection
A: CloudWatch Alarms (Composite)
Route
Amazon SNS
P+E: Remediation Lambda
Loop 2: Infrastructure Healing (MAPE-K)
Actions: Scale memory, provisioned concurrency, model fallback, config update
Monitor
Monitor
Update config
Log action
Shared Knowledge Base (K)
User Query
Response
Request
API Gateway
Query
Monitor
Response
Query Lambda
Amazon S3 (Source Docs)
Ingest
Core RAG Pipeline
Bedrock Knowledge Bases
Amazon Bedrock RetrieveAndGenerate
Response
Retrieve
Embed via Titan Embeddings v2
Generate
OpenSearch Serverless (Vector Store)
Claude 3 Sonnet (Generation)
Re-sync
Pass
Healing Complete
Log result


EventBridge
Scheduled Rule
(every 6 hrs)
Trigger
M: Monitor Lambda
Scrape sources,
SHA-256 hash,
compare metadata
Drift
etected
alyze
d chunks,
distance,
ld check
Plan
e vs. full
lexing
oop 1: Knowledge Healing (MAPE-K)
bdaRe-chunk, re-
bed,
vectors
Read/Write
metadata
Validate
RAGAS Validation
(faithfulness check)
Fail: Rollback
& Alert
Log action
Human
Operator
DynamoDB
Healing history,
source metadata,
thresholds,
audit trail